

Linear programming as a tool for the optimal control of invasive species

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Biological invasions and control

- Invasive spread of alien species a widespread and costly ecological problem
- Need to design effective control strategies subject to budget constraints





What is the objective of control?

- Minimize extent of invasion?
- Eliminate the invasive at minimal cost?
- Minimize environmental impact of the invasive?

How do we calculate the optimal strategy anyway?



Talk outline

- Show how optimal control of invasions can be solved using linear programming algorithms
- optimal removal of a stage-structured invasive
- effect of economic discounting
- optimal control of an invasive which damages its environment

Linear Programming

- Technique for finding optimal solutions to linear control problems
- Fast and efficient compared with other computationally intensive optimization methods
- Assumes that in early stages of invasion, growth is approximately exponential



Model system: invasive *Spartina*

- Introduced to Willapa Bay, WA c. 100 years ago
- Annual growth rate approx 15%; occupies 72 sq km
- Reduces shorebird foraging habitat...
- and changes tidal height



Model system: invasive *Spartina*



Seedling



Isolate

Rapid growth (asexual)
Highest reproductive value



Meadow

High seed production (sexual)
Highest contribution to next generation



Mathematical model

N_t = population in year t

$$N_{t+1} = L (N_t - H_{t+1})$$

H_t = area removed in year t

L = population growth matrix



$$N_T = L^T N_0 - \sum_{t=1}^T L^{T+1-t} H_t$$

linear in control variables



Optimization problem

Objective: minimize population size after T years of control

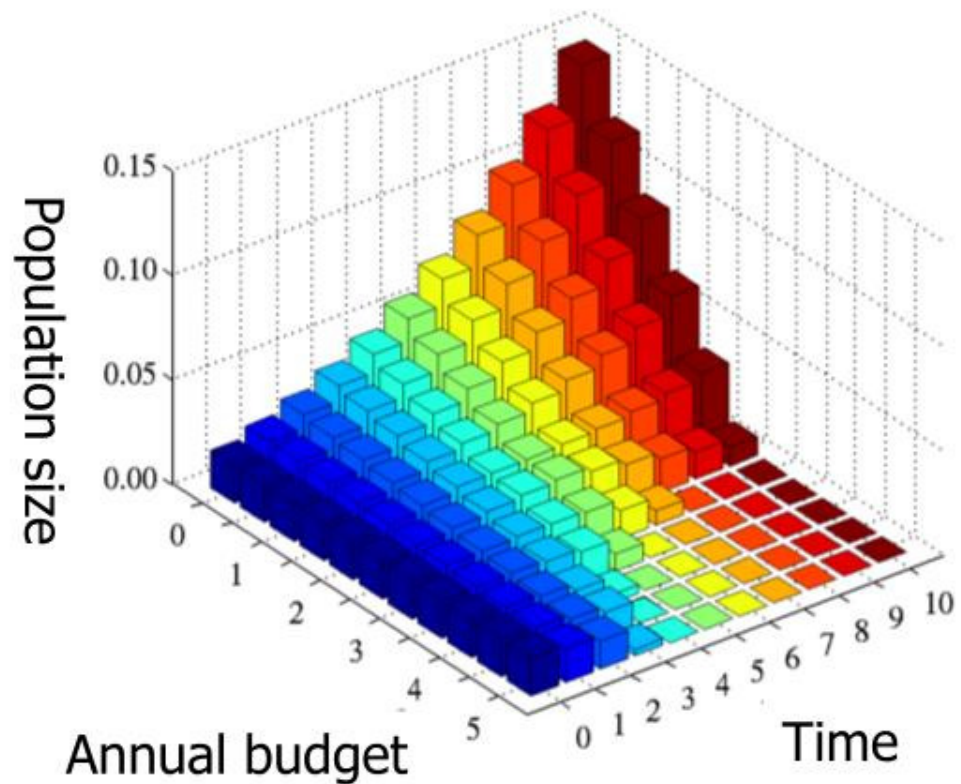
Constraints

Non-negativity: $H_{t,j}, N_{t,j} \geq 0$

Budget: $c_H \cdot H_t < C$

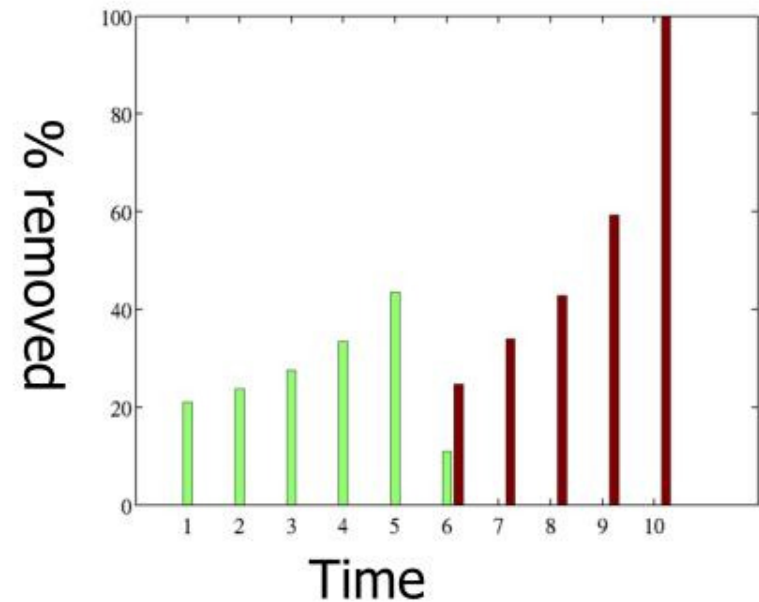
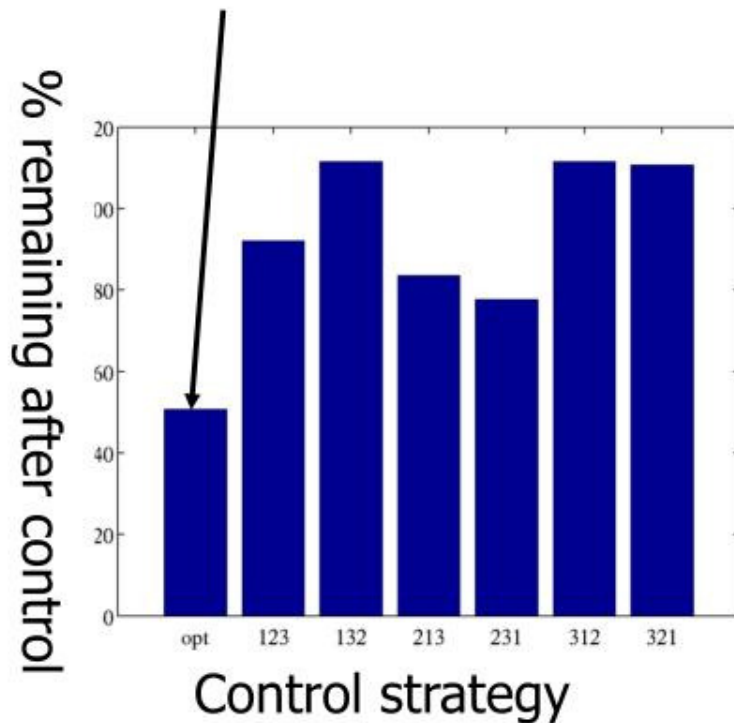
Results

Sufficient annual budget crucial to success of control



Results

Optimal strategy really is optimal!



Shift from removing isolates to meadows



Effect of discounting

Goal: eliminate population by time T at minimal cost

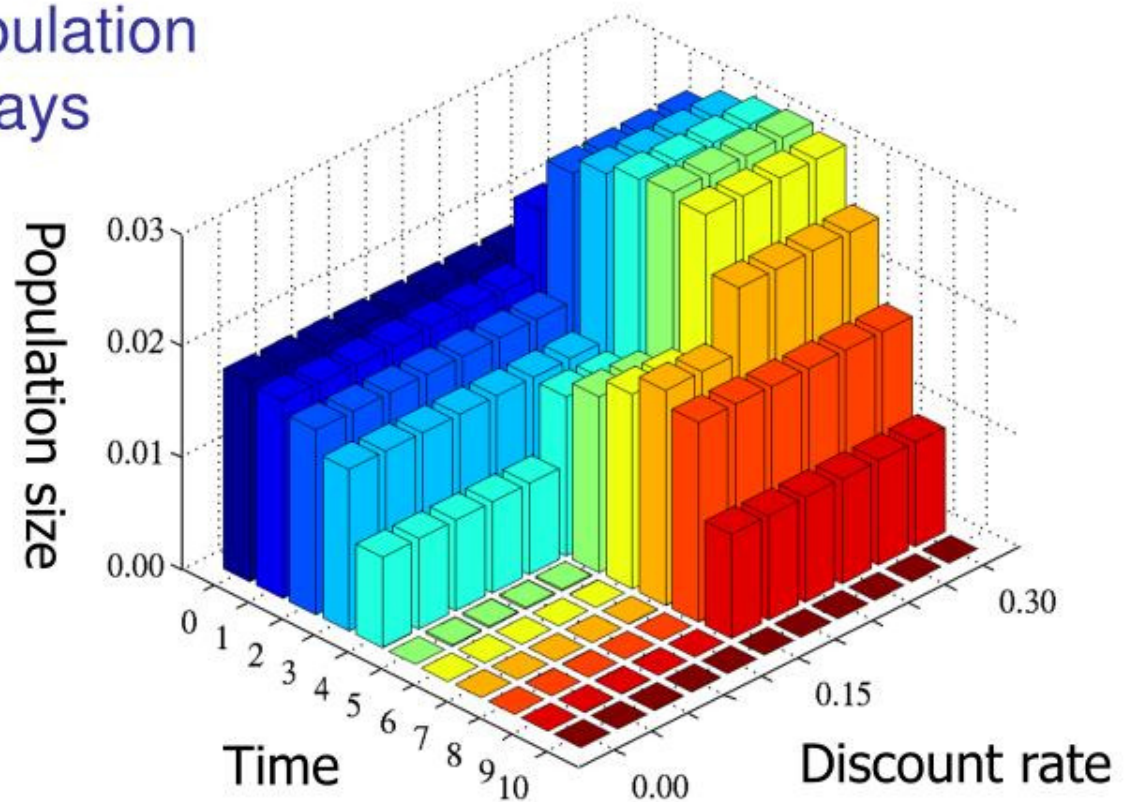
Objective: Minimize total cost of control subject to discounting at rate γ

$$\text{i.e. } \sum_{t=1}^T c_H \cdot H_t e^{-\gamma t}$$

Constraints : same as before, but now population in time T must be zero

Effect of discounting

As discount rate approaches population growth rate, it pays to wait





Adding damage and restoration

- Area from which invasive is removed remains damaged ($H_t \rightarrow D_t$)
- This damage can be controlled through restoration or mitigation ($D_t \rightarrow R_t$)
- Proportion **1-P** of damaged area recovers naturally each year

Model:
$$N_{t+1} = L (N_t - H_{t+1})$$

$$D_{t+1} = P (D_t + H_{t+1} - R_{t+1})$$



Optimization problem

Objective: minimize total cost of invasion

Removal cost $\sum_{t=1}^T c_H \cdot H_t e^{-\gamma t}$



Optimization problem

Objective: minimize total cost of invasion

Removal cost $\sum_{t=1}^T c_H \cdot H_t e^{-\gamma t}$

Restoration cost $\sum_{t=1}^T c_R \cdot R_t e^{-\gamma t}$



Optimization problem

Objective: minimize total cost of invasion

Removal cost $\sum_{t=1}^T c_H \cdot H_t e^{-\gamma t}$

Restoration cost $\sum_{t=1}^T c_R \cdot R_t e^{-\gamma t}$

Environmental cost $\sum_{t=1}^T c_E \cdot (N_t + D_t) e^{-\gamma t}$



Optimization problem

Objective: minimize total cost of invasion

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Restoration cost $\sum_{t=1}^T c_R \cdot R_t e^{-\gamma t}$

Environmental cost $\sum_{t=1}^T c_E \cdot (N_t + D_t) e^{-\gamma t}$

Salvage cost $c_H \cdot N_T + \sum_{t=T}^{\infty} c_E \cdot P^{T-t} (N_T + D_T) e^{-\gamma t}$



Optimization problem

Objective: minimize total cost of invasion

Removal cost $\sum_{t=1}^T c_H \cdot H_t e^{-\gamma t}$

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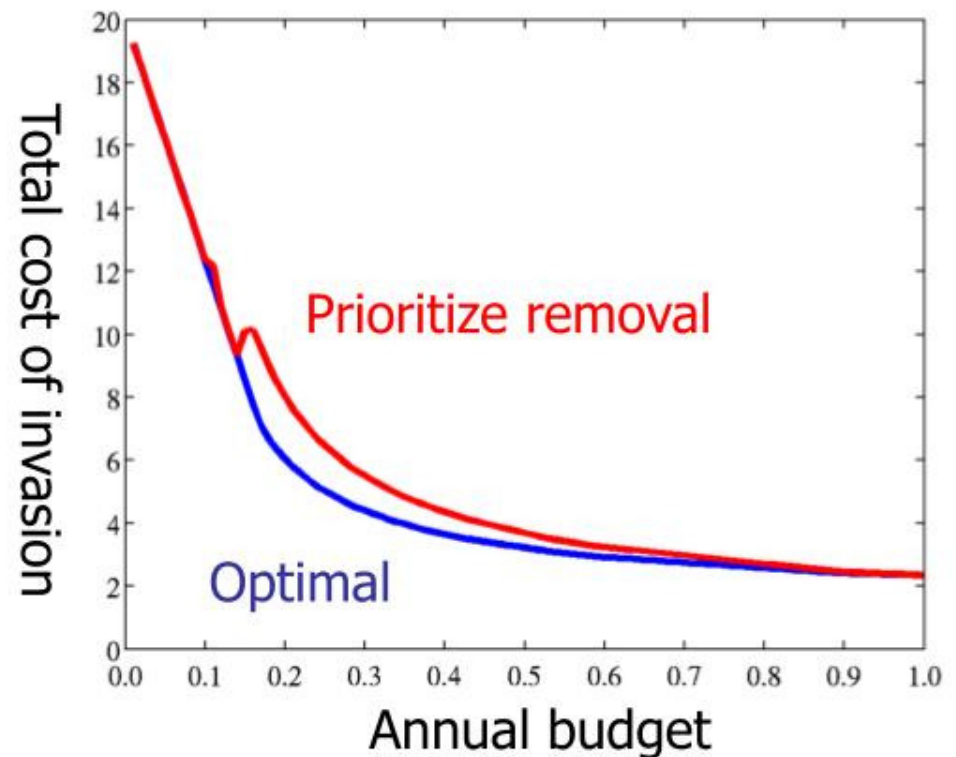
Salvage cost $c_H \cdot N_T + \sum_{t=T}^{\infty} c_E \cdot P^{T-t} (N_T + D_T) e^{-\gamma t}$

Constraints: non-negativity of variables

Annual budget: $c_H \cdot H_t + c_R \cdot R_t \leq C$

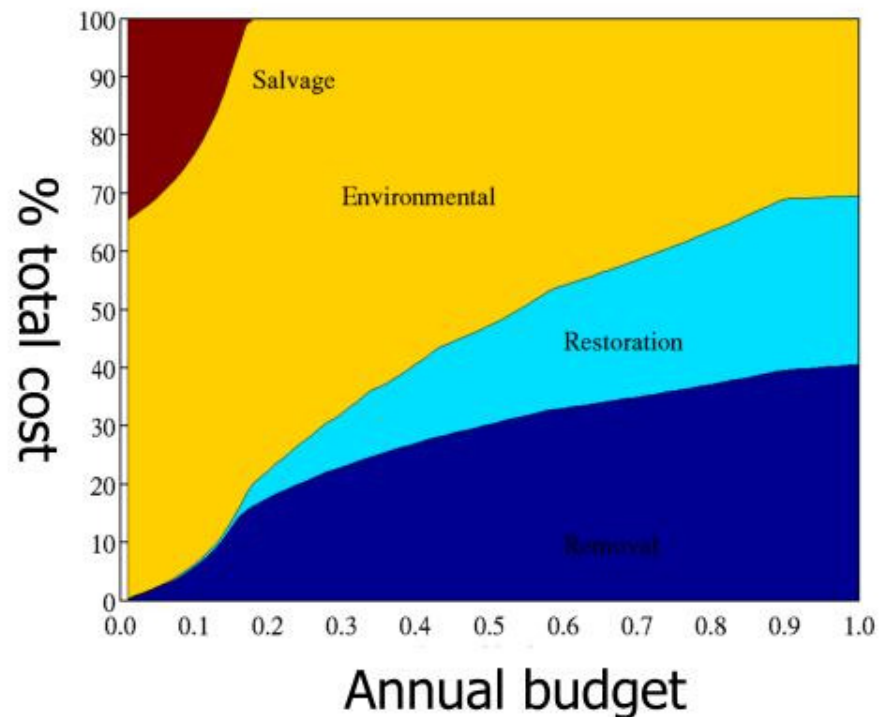
Results

Optimal strategy always better than prioritizing removal over restoration



Results

- Salvage cost
- Environmental cost
- Restoration cost
- Removal cost



Only restore when budget is sufficient to eliminate invasive



Summary

- Linear programming is a fast, efficient method for calculating optimal control strategies for invasives
- Changing which stage class is prioritized by control is often optimal
- The degree of discounting affects the timing of control
- If annual budget high enough, investing in restoration reduces total cost of invasion

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Maybe I should
just stick to
modeling...

THANKS FOR LISTENING!

